BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Timed Paper 2 Practice

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. In a file-size calculation, should the final answer include units?

2. Q = (A AND B) OR C. A=1, B=1, C=0. What is Q?

A) 1 B) 0
C) 2 D) 10

3. "A translator is needed only to change the program's comments into another human language." Correct this. Explain why.

4. Merge the sorted lists [2, 7] and [4, 6] into ascending order. Give the values in order. Why not just join the lists?

Today's objectives

Explain the core ideas of timed paper 2 practice using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Paper 2 timing
The OCR J277/02 paper has 80 marks in 90 minutes.

Code tracing
Write a reliable value table before selecting the output.

Program design
Plan input, processing, selection, iteration and output.

Error spotting
Check indexes, loop bounds, equality tests and branching.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

Paper 2 timing	The OCR J277/02 paper has 80 marks in 90 minutes.
Code tracing	Write a reliable value table before selecting the output.
Program design	Plan input, processing, selection, iteration and output.
Error spotting	Check indexes, loop bounds, equality tests and branching.

I Do - One worked example

Trace count=0; for i=1 to 5; if i MOD 2=0 then count=count+1. Output count.

1. Loop through i=1,2,3,4,5.
2. Only i=2 and 4 satisfy even test.
3. Count progresses 0,1,1,2,2.
4. Final output is 2.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Design a three-line correction to a failed validation check using OR.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Count the 1 bits in a five-bit binary value using iteration and selection.

```
array bits = [1,0,1,1,0]
ones = 0
for i = 0 to 4
if bits[i] == 1 then
ones = ones + 1
endif
next i
print(ones)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State what an accumulator stores. [2 marks]

Q3. Write a valid condition for 0<=score<=100. [2 marks]

Q4. Explain one common index error in zero-based arrays. [3 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A trace table helps follow loop values and decisions. ☐ T ☐ F
2. The OCR J277/02 paper has 80 marks in 90 minutes. ☐ T ☐ F
3. Write a reliable value table before selecting the output. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Yes; show a calculation and label the final unit, such as bits or bytes. ☐

Q2 A - 1 ☐

Q3 A CPU runs machine instructions. High-level source code must be translated before or as it runs. ☐

Q4 (a) [2, 4, 6, 7]. Keep taking the smaller next item. Just joining the lists would put 7 before 4. ☐

Code First - Model

The original five-bit number 10110 has three 1 bits, so output is 3. Changing index 1 to 1 gives four 1 bits (11110). The original binary value is 16+4+2 = 22 denary; counting set bits and converting place values are different operations.

We Do - Check the model

1. Check endpoints. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 A running total updated as a program executes. ☐

Q3 score >= 0 AND score <= 100. ☐

Q4 Using length instead of length-1 as final index can access outside array bounds. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: